

Week 6 Report — Testing & System Integration

Programme: IIMSTC – IoT & Embedded Systems (Cohort 4)

Team Size: 15 Members

Hardware Team	App Development Team	Software Team
Harsha Vardhan S	Krithi	Vijetha
Ananya Chandrashekar	Nagaraju M	S Vidyashree
Palaparthi Krishna Vamsi	Charan B S	Sachidananda M M
Durga Prasad S	—	—
Ramitha R	—	—
Hadiya Khan	—	—
C Pavitra	—	—
Abhinav P	—	—

Day 1 — Mobile App Integration with ESP32

The mobile application was integrated with the ESP32 module. Data synchronization between the hardware device and the app was tested, ensuring real-time updates and proper display of temperature and system status.

Day 2 — ESP32 & Website Connectivity Testing

The ESP32 was connected to the web platform, and initial testing was performed to verify data transmission. Sensor readings were successfully sent to the website, ensuring proper communication between hardware and frontend interface.

Day 3 — API Communication Testing

Backend APIs were tested for stable data handling and communication after integrating the website and mobile application with the ESP32. Data exchange between ESP32, website, and mobile application was verified. Issues related to latency and data consistency were identified and resolved.

Day 4 — Full System Integration

Complete system integration was carried out by combining hardware (ESP32, sensors,

actuators), software (backend), and user interfaces (website and mobile app). End-to-end testing ensured seamless interaction across all components.

Day 5 — System Testing & Debugging

Comprehensive testing was performed under different conditions to validate system performance. Bugs related to data delay, incorrect readings, and interface glitches were fixed. System reliability and response time were improved.

Day 6 — Documentation & Report Preparation

All project work was documented, including system architecture, circuit design, software flow, and integration process. The final report was prepared with proper formatting, detailing the implementation and testing outcomes.

Summary

Week 6 focused on testing, integration, and documentation of the complete system. Successful communication between ESP32, website, and mobile application was achieved. The system was validated through thorough testing, and all components were unified into a fully functional IoT solution, ready for demonstration and deployment.